

Classwork Problems

Unit 2: Differentiation and its Applications

L11: Differentiation Recap – Problems

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|---|--|
| (1) $y = (x+2)^2$ | Ans. $2x + 4$ |
| (2) $y = e^{2x+5}$ | Ans. $2e^{2x+5}$ |
| (4) $y = (2x-3)^5$ | Ans. $10(2x-3)^4$ |
| (5) $y = \cos(e^{\sin x})$ | Ans. $-\sin(e^{\sin x}) \cdot e^{\sin x} \cdot \cos x$ |
| (6) $y = \sqrt{3^x - 1}$ | Ans. $\frac{3^x \log 3}{2\sqrt{3^x - 1}}$ |
| (7) $y = \sec^2(4x-3)$ | Ans. $8\sec^2(4x-3) \cdot \tan(4x-3)$ |
| (8) $y = \tan^4(x^3+2)$ | Ans. $12x^2 \tan^3(x^3+2) \sec^2(x^3+2)$ |
| (9) $y = \frac{1}{\sqrt[3]{\sin^2 4x}}$ | Ans. $-\frac{8 \cos 4x}{3\sqrt[3]{\sin^5 4x}}$ |
| (10) $y = x^2 \cos^2 3x$ | Ans. $-6x^2 \cos 3x \cdot \sin 3x + 2x \cos^2 3x$ |

L12: Derivatives of inverse trigonometric functions

<u>Expression</u>	<u>Substitution</u>
$\sqrt{a^2 - x^2}$	$x = a \sin \theta \text{ or } a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta \text{ or } a \cot \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta \text{ or } a \operatorname{cosec} \theta$
$\sqrt{a-x} \text{ and } \sqrt{a+x}$	$x = a \cos \theta \text{ or } a \cos 2\theta$
$1 - 2x^2$	$x = \sin \theta$
$2x^2 - 1$	$x = \cos \theta$
$\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}, \frac{2x}{1-x^2}$	$x = \tan \theta$

Find $\frac{dy}{dx}$ for the following

(1) $y = \cos^{-1}(1-2x^2)$ Ans. $\frac{2}{\sqrt{1-x^2}}$

Solution: $y = \cos^{-1}(1-2x^2)$

Put $x = \sin \theta$

$\therefore y = \cos^{-1}(1-2\sin^2 \theta)$

$\therefore y = \cos^{-1}(\cos 2\theta)$

$\therefore y = 2\theta$

$\therefore y = 2\sin^{-1} x$

Differentiate w.r.t. x

$$\frac{d}{dx}(y) = \frac{d}{dx}(2 \sin^{-1} x)$$

$$\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$$

$$(2) y = \sin(\cos^{-1} x)$$

$$\text{Ans. } \frac{-x}{\sqrt{1-x^2}}$$

$$(3) y = \tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$$

$$\text{Ans. } \frac{1}{2}$$

$$(4) y = \operatorname{cosec}^{-1} \sqrt{1+x^2}$$

$$\text{Ans. } -\frac{1}{1+x^2}$$

$$(5) y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$\text{Ans. } \frac{2}{1+x^2}$$

$$(6) y = \tan^{-1} \sqrt{\frac{1-x}{1+x}}$$

$$\text{Ans. } -\frac{1}{\sqrt{1-x^2}}$$

$$(7) y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$\text{Ans. } \frac{2}{1+x^2}$$

$$(8) y = \sec^{-1}\left(\frac{1}{2x^2-1}\right)$$

$$\text{Ans. } -\frac{2}{\sqrt{1-x^2}}$$

$$(9) y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$$

$$\text{Ans. } \frac{3}{1+x^2}$$

L13, 14: Derivatives of exponential and logarithmic functions

Find the derivative of the following

$$(1) y = x e^x$$

$$\text{Ans. } x e^x + e^x$$

Solution: $y = x e^x$

Differentiate w.r.t. x

$$\frac{d}{dx}(y) = \frac{d}{dx}(x e^x)$$

$$\frac{dy}{dx} = x \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(x)$$

$$\frac{dy}{dx} = x e^x + e^x$$

$$(2) y = 2^x$$

$$\text{Ans. } 2^x \log 2$$

$$(3) y = e^{\sin^{-1} x}$$

$$\text{Ans. } \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$$

$$(4) y = e^{x^3}$$

$$\text{Ans. } 3x^2 e^{x^3}$$

$$(5) y = e^x \cdot 4^x$$

$$\text{Ans. } e^x 4^x \log 4 + 4^x e^x$$

$$(6) y = \cos^{-1}(e^x)$$

$$\text{Ans. } \frac{-e^x}{\sqrt{1-e^{2x}}}$$

$$(7) \cos(\log x + e^x)$$

$$\text{Ans. } -\sin(\log x + e^x) \cdot \left(\frac{1}{x} + e^x\right)$$

$$(8) y = \frac{e^x}{\sin x}$$

$$(9) y = \log(\cos e^x)$$

$$(10) y = \log(\log x)$$

$$\text{Ans. } e^x \cdot \frac{(\sin x - \cos x)}{\sin^2 x}$$

$$\text{Ans. } -e^x (\tan e^x)$$

$$\text{Ans. } \frac{1}{x \cdot \log x}$$

Logarithmic differentiation

A function in the form $y = f(x)^{g(x)}$ cannot be differentiated directly because it is different from $y = f(x)^n$. In such cases we take logarithm on both sides, simplify the R.H.S and then differentiate both sides w.r.t x and obtain $\frac{dy}{dx}$. Such type of differentiation is called logarithmic differentiation.

eg:

$$y = x^x$$

$$\log y = \log(x^x)$$

$$\log y = x \log x$$

Differentiate w.r.t x

$$\frac{1}{y} \frac{dy}{dx} = x \times \frac{1}{x} + \log x \times 1$$

$$\frac{dy}{dx} = y[1 + \log x]$$

$$\frac{dy}{dx} = x^x(1 + \log x)$$

Find $\frac{dy}{dx}$ for the following

$$(1) y = \log(x^2 + 1)$$

$$\text{Ans. } \frac{2x}{x^2 + 1}$$

Solution: $y = \log(x^2 + 1)$

Differentiate w.r.t. x

$$\frac{d}{dx}(y) = \frac{d}{dx}(\log(x^2 + 1))$$

$$\frac{dy}{dx} = \frac{1}{x^2 + 1} \frac{d}{dx}(x^2 + 1)$$

$$\frac{dy}{dx} = \frac{2x}{x^2 + 1}$$

$$(2) y = x^x$$

$$\text{Ans. } x^x(1 + \log x)$$

$$(3) y = (\sin x)^x$$

$$\text{Ans. } (\sin x)^x (x \cdot \cot x + \log(\sin x))$$

$$(4) y = (x)^{\sin x} + x^{(1/x)}$$

$$\text{Ans } x^{\sin x} \left(\frac{\sin x}{x} + \cos x \cdot \log x \right) + x^{1/x} \left(\frac{1 - \log x}{x^2} \right)$$

$$\begin{aligned}
(5) \ x^y &= e^{x-y} & \text{Ans. } \frac{\log x}{(1+\log x)^2} \\
(6) \ \sin y &= \log(x+y) & \text{Ans. } \frac{1}{(x+y)\cos y - 1} \\
(7) \ y &= \frac{e^x \cdot \sin x}{x^3} & \text{Ans. } \frac{e^x \cdot \sin x}{x^3} \left(1 + \cot x - \frac{3}{x} \right) \\
(8) \ y &= 5^x \cdot \sec x \cdot \tan 2x & \text{Ans. } 5^x \cdot \sec x \cdot \tan 2x (\log 5 + \tan x + 4 \operatorname{cosec} 4x) \\
(9) \ y^x &= x^y & \text{Ans. } \frac{y}{x} \left(\frac{y-x \log y}{x-y \log x} \right) \\
(10) \ y &= (1+x)(1+x^2)(1+x^4)(1+x^8) \\
\text{Ans. } & (1+x)(1+x^2)(1+x^4)(1+x^8) \left[\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8} \right]
\end{aligned}$$

L15: Derivatives of implicit functions

If a variable y can be expressed as a function of another variable x i.e $y = f(x)$ then the function $f(x)$ is called the **explicit function** of x .

$$\text{eg: } y = x^2 + x - 3, \quad y = \sin 3x + e^x$$

If neither x nor y can be expressed explicitly as a function of other variable then such a function is called **implicit function**.

$$\text{eg: } \log(x+y) = \sin(x+y), \quad x^m y^n = (x+y)^{m+n}$$

- In order to differentiate an implicit function, we have to differentiate the terms in x and y w.r.t x simultaneously.
- Further to differentiate a term in y w.r.t x , we differentiate it similar to the corresponding term in x and multiply by $\frac{dy}{dx}$ using chain rule.

Find $\frac{dy}{dx}$ for the following:

$$(1) \ x^3 + y^3 = 3axy \quad \text{Ans. } \frac{ay - x^2}{y^2 - ax}$$

$$\text{Solution: } x^3 + y^3 = 3axy$$

Differentiate w.r.t. x

$$\frac{d}{dx}(x^3 + y^3) = \frac{d}{dx}(3axy)$$

$$3x^2 + 3y^2 \frac{dy}{dx} = 3a \left[x \frac{dy}{dx} + y \right]$$

$$(y^2 - ax) \frac{dy}{dx} = ay - x^2$$

$$\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

$$(2) xy^2 = x^3 + y$$

$$\text{Ans. } \frac{3x^2 - y^2}{2xy - 1}$$

$$(3) x^4 + 5x^2y = 8xy^2 - 6y^3$$

$$\text{Ans. } \frac{8y^2 - 4x^3 - 10xy}{5x^2 - 16xy + 18y^2}$$

$$(4) \sin^2 y + \cos xy = 2$$

$$\text{Ans. } \frac{y \sin xy}{\sin 2y - x \sin xy}$$

$$(5) \sin^2 x + \cos^2 y = 1$$

$$\text{Ans. } \frac{\sin 2x}{\sin 2y}$$

$$(6) y - \tan^{-1}y = x$$

$$\text{Ans. } \frac{1}{y^2} + 1$$

$$(7) 2xy = 1$$

$$\text{Ans. } \frac{-y}{x}$$

$$(8) \sin xy = y$$

$$\text{Ans. } \frac{y \cos xy}{1 - x \cos xy}$$

$$(9) \sqrt{x} - \sqrt{y} = \sqrt{3}$$

$$\text{Ans. } \frac{\sqrt{y}}{\sqrt{x}}$$

$$(10) \frac{1}{x} + \frac{1}{y} = \frac{1}{2}$$

$$\text{Ans. } \frac{-y^2}{x^2}$$

L16: Derivatives of parametric functions

If $x = f(t)$ and $y = g(t)$ are differentiable functions of parameter t , then y is a differentiable function of x and

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}, \quad \frac{dx}{dt} \neq 0$$

Eg: Find $\frac{dy}{dx}$ for the following:

$$(1) x = at^2; y = 2at$$

$$\text{Ans. } \frac{1}{t}$$

Solution: $x = at^2; y = 2at$

Differentiate w.r.t. t

$$\frac{d}{dt}(x) = \frac{d}{dt}(at^2); \frac{d}{dt}(y) = \frac{d}{dt}(2at)$$

$$\frac{dx}{dt} = 2at; \frac{dy}{dt} = 2a$$

$$\text{We know that, } \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{dy}{dx} = \frac{2a}{2at}$$

$$\frac{dy}{dx} = \frac{1}{t}$$

(2) $x = \sin(\log t); y = \log(\sin t)$	Ans. $\frac{t \cdot \cos t}{\cos(\log t)}$
(3) $x = \cos t; y = \sin t$ at $t = \frac{\pi}{4}$	Ans. -1
(4) $x = a(\cos \theta + \theta \sin \theta); y = a(\sin \theta - \theta \cos \theta)$	Ans. $\tan \theta$
(5) Differentiate $\tan^{-1} x$ w.r.t. $\frac{1}{1+x^2}$.	Ans. $-\frac{1+x^2}{2x}$
(6) $x = (\cos \theta - \cos 2\theta); y = (\sin \theta - \sin 2\theta)$	Ans. $\frac{\cos \theta - 2 \cos 2\theta}{2 \sin 2\theta - \sin \theta}$
(7) $x = 4t^2; y = \frac{4}{t^2}$	Ans. $\frac{-1}{t^4}$
(8) $x = a \sec \theta; y = b \tan \theta$	Ans. $\frac{b}{a} \operatorname{cosec} \theta$

Additional problems

Find $\frac{dy}{dx}$ if

- 1) $y = (2x + 5)^3$ 2) $y = \sqrt{\sin x}$ 3) $y = \cos^4 x$ 4) $y = e^{\tan x}$ 5) $y = \log(\sec x)$
 6) $y = \sin mx$ 7) $y = \sec \sqrt{x}$ 8) $y = \cos(2 - 3x)$

Differentiate the following w.r.t 'x'.

- 1) $(2x^2 - 3x + 1)^3$ 2) $\cos(e^{5x})$ 3) $e^{\sin^2 x}$ 4) $\log(\sec x + \tan x)$ 5) $y = \operatorname{cosec}^3(5x+1)$

Find $\frac{dy}{dx}$ if

- (i) $y = \sin^{-1} \sqrt{x}$
 (ii) $y = \cos^{-1}(2x)$
 (iii) $y = \tan^{-1}(\log x)$

Find $\frac{dy}{dx}$ if

- (i) $y = \sin^{-1}(\cos x)$ (ii) $y = \cos^{-1}(2x + 3)$ (iii) $y = \cos^{-1}(2x^2 - 1)$ (iv) $y = \sin^{-1}(3x - 4x^3)$

(i) $y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$

(ii) $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$

(iii) $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$

(iv) $y = (1+x^2) \tan^{-1} x$

1. Find $\frac{dy}{dx}$ if

- (i) $x^2 + y^2 + 2y = 0$ (ii) $x^2 + y^2 = a^2$ (iii) $xy = c^2$ (iv) $\sqrt{y} + \sqrt{x} = \sqrt{a}$

$$(i) \cos x + \sin y = c \quad (ii) x^2 \sin y = c \quad (iii) y = a + xe^y$$

2. Find $\frac{dy}{dx}$ if

$$(i) x^3 + y^3 = 3axy \quad (ii) x^2 + y^2 - 4x + 6y - 5 = 0 \quad (iii) x^2 + y^2 = xy$$

$$(iv) ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$(i) x^3 + y^3 = 3axy$$